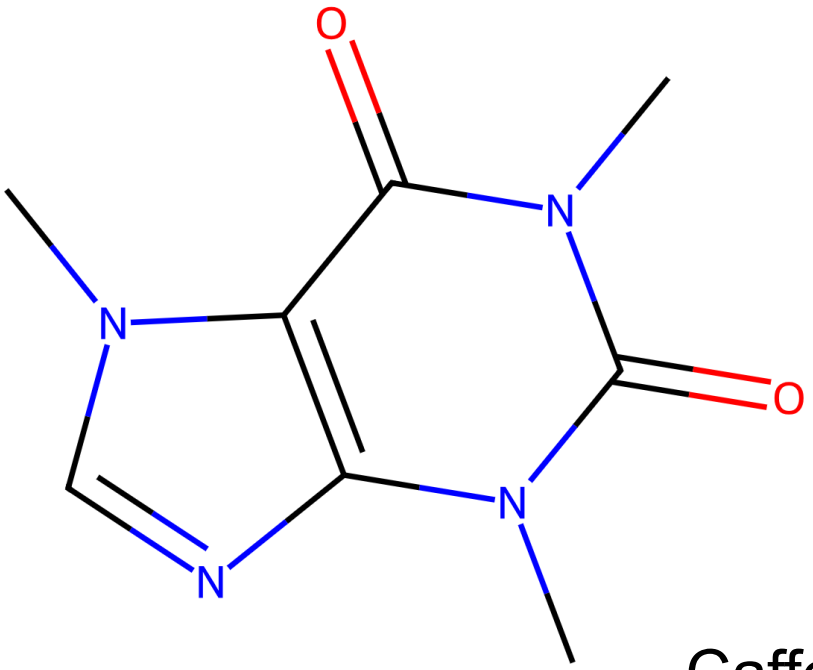
Chapter 2: From Molecules to Features – Preparing Training Data

Chem 361

Xuhui Huang

***"A mathematician is a machine that converts coffee into theorems."* - Paul Erdős**



Caffeine

Can we represent chemical compounds through mathematical and machine-readable descriptions?

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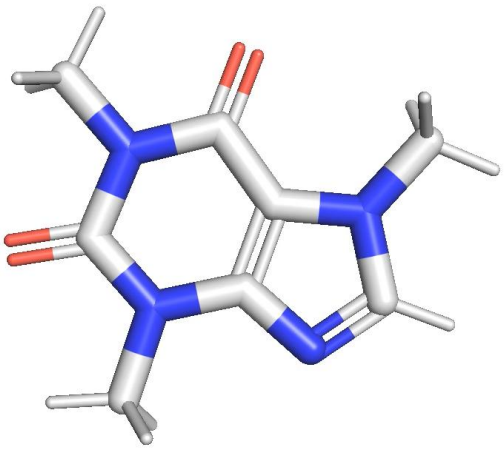
Features of Chemical Compounds



Caffeine

Generic names	Caffeine, Guaranine, 1,3,7-Trimethylxanthine
Molecular Weight	194.19
# of H-Bond Donors	0
# of H-Bond Acceptors	6
# of Rotable Bonds	0
logP (octanol-water partition coefficient)	-1.03
Molecular Formula	$C_8H_{10}N_4O_2$
SMILES	Cn1c(=O)c2c(ncn2C)n(C)c1=O
InChI (International Chemical Identifier)	InChI=1S/C8H10N4O2/c1-10-4-9-6-5(10)7(13)12(3)8(14)11(6)2/h4H,1-3H3

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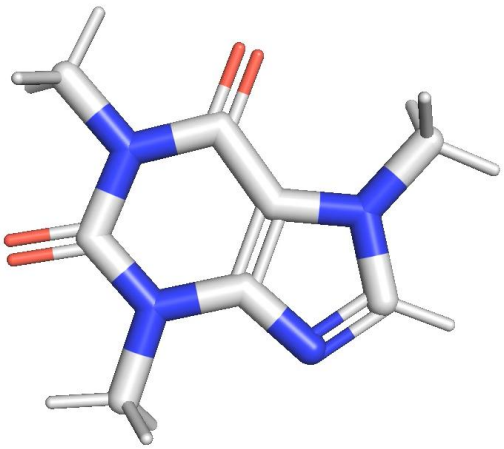
Features of Chemical Compounds



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Molecular Properties

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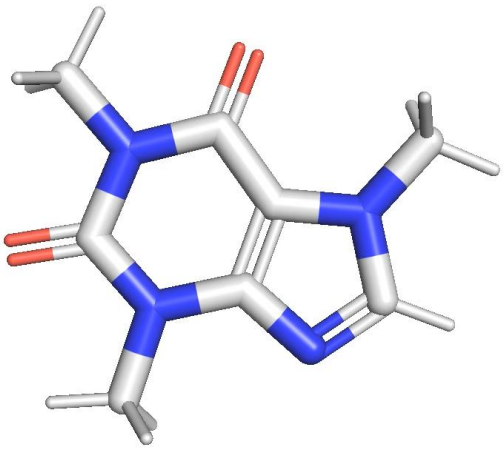
Features of Chemical Compounds



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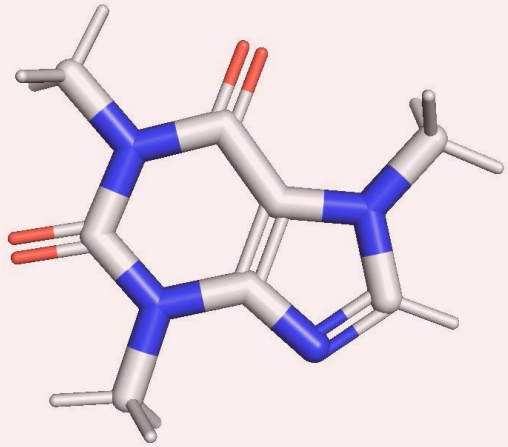
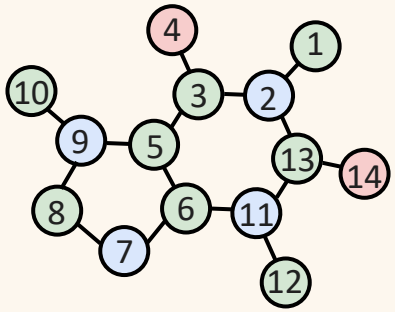
Features of Chemical Compounds

3D coordinates





Graph

Generic names	Caffeine, Guaranine, 1,3,7-Trimethylxanthine
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Outline

- **Molecular property descriptors**
- String representations
- Molecular graphs
- 3D-representations
- Binding environment features
- Bottom-up representations for peptides

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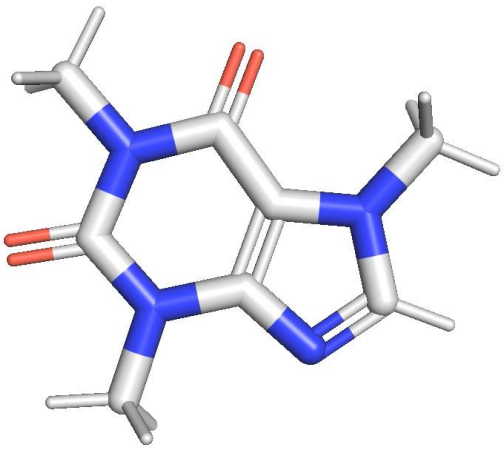
Features of Chemical Compounds



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Molecular Properties

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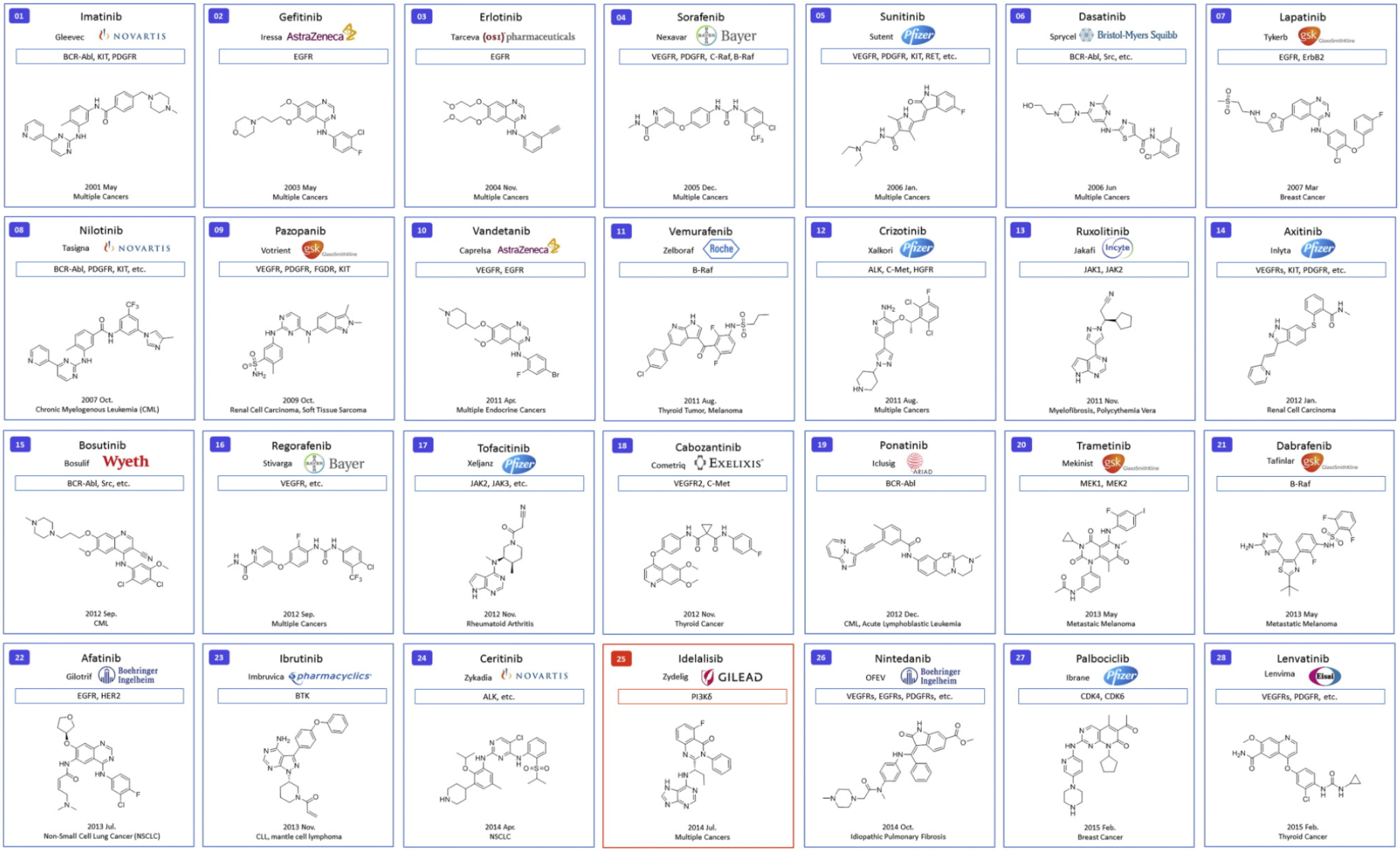
Drug-likeness Molecular Properties



Trends in Pharmacological Sciences 36, 422-439, (2015)

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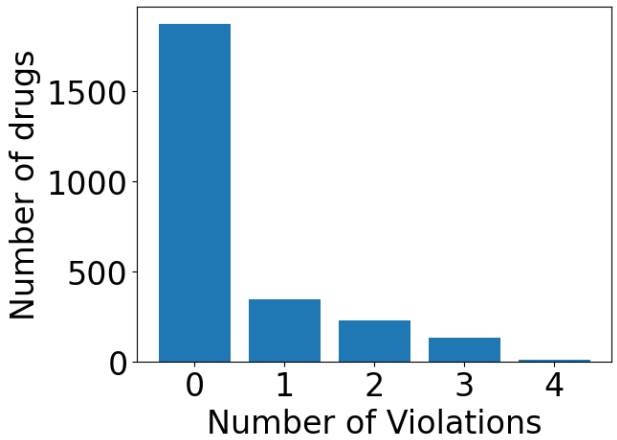
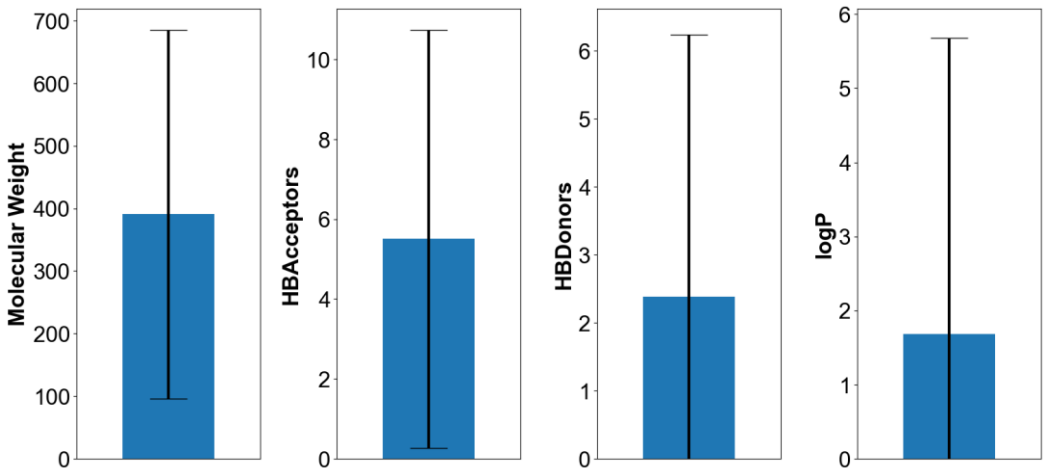
Drug-likeness Molecular Properties

Lipinski's rule of five (RO5): a rule of thumb to evaluate drug-likeness of chemical compounds:

1. Molecular weight <= 500 daltons
2. # of hydrogen bond donors <= 5
3. # of hydrogen bond acceptors <= 10
4. LogP (octanol-water partition coefficient) <= 5.0





Most FDA approved drug compounds violate 0 or 1 rule!

Data: 2,584 drug compounds from DrugBank

https://go.drugbank.com/

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Drug Bank Database

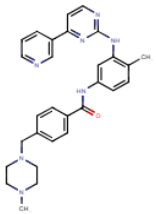
Time magazine, 2001

“Wonder drug of century”

In Phase I clinic trial, *100%* patients experienced completed remission of Leukemia (blood counts return to normal).

NEJM, 344, 1031, (2001)

DRUGBANK

Brand Names	*Gleevec, Glivec, Imkeldi*
Generic Name	Imatinib
DrugBank Accession Number	DB00619
Type	Small Molecule
Groups	Approved
Structure	 3D Download Similar Structures
Weight	Average: 493.6027 Monoisotopic: 493.259008649
Chemical Formula	$C_{29}H_{31}N_7O$

https://go.drugbank.com/

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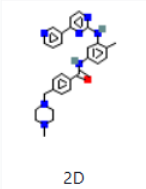
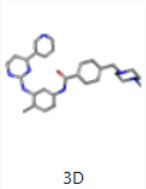
Pub Chem Database

PubChem

Imatinib

PubChem CID	5291
Structure	 2D  3D
Chemical Safety	Irritant Health Hazard Laboratory Chemical Safety Summary (LCSS) Datasheet
Molecular Formula	$C_{29}H_{31}N_7O$
Synonyms	Imatinib 152459-95-5 STI571 Imatinib (STI571) sti-571 View More...
Molecular Weight	493.6 g/mol *Computed by PubChem 2.2 (PubChem release 2021.10.14)*

CONTENTS

Title and Summary
1 Structures
2 Names and Identifiers
3 Chemical and Physical Properties
4 Spectral Information
5 Related Records
6 Chemical Vendors
7 Drug and Medication Information
8 Pharmacology and Biochemistry
9 Use and Manufacturing
10 Safety and Hazards
11 Toxicity
12 Associated Disorders and Diseases
13 Literature
14 Patents
15 Interactions and Pathways
16 Biological Test Results
17 Classification
18 Information Sources

Launched in 2004 and currently contains >119.3 million unique chemical structures.

https://pubchem.ncbi.nlm.nih.gov/

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RDKit: Open-Source Cheminformatics Software

Print out molecular properties of Imatinib:

```python
import rdkit.Chem as Chem
from rdkit.Chem import Descriptors

descriptors = {
    "Molecular Weight": Descriptors.MolWt,
    "HDonors": Chem.Lipinski.NumHDonors,
    "HAcceptors": Chem.Lipinski.NumHAcceptors,
    "Number of Rings": Descriptors.RingCount,
    "logP": Descriptors.MolLogP,
    "Rotable Bonds": Descriptors.NumRotatableBonds,
}

for desp_name, desp_fn in descriptors.items():
    desp_val = desp_fn(mol)
    print(f"{desp_name}: {desp_val}")
```

✓ 0.0s

```
Molecular Weight: 493.61500000000024
HDonors: 2
HAcceptors: 7
Number of Rings: 5
logP: 4.590320000000003
Rotable Bonds: 7
```

Compute drug-likeness score (QED) and synthetic availability score (SA score):

```python
# quantitative estimation of drug-likeness (QED)
q = Descriptors.qed(mol)
print("QED:", q)

import os, sys
from rdkit.Chem import RDConfig
# using literature contributors
# from https://github.com/rdkit/rdkit/tree/master/Contrib
sys.path.append(os.path.join(RDConfig.RDContribDir, 'SA_Score'))
import sascorer
s = sascorer.calculateScore(mol)
print("Synthetic Availability:", s)
```

✓ 0.0s

```
QED: 0.38941616262964607
Synthetic Availability: 2.3316597651325157
```

More exercise of using RDKit is available in homework #1.

https://www.rdkit.org/

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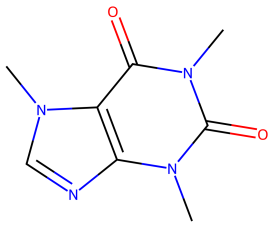
Fingerprints

Convert each chemical compound to an array by pre-defining a list of properties or structural information of interest



Caffeine

MW	HBA	HBD	logP
194.2	6	0	-1.03

molecular properties (Real numbers)

-R	-OH	-O-	$-NH_2$	-CONH-	-CHO	-COOH
1	0	0	0	0	0	0

Functional groups (Binary)

3	0	0	0	0	0	0

Functional groups (Count-based)

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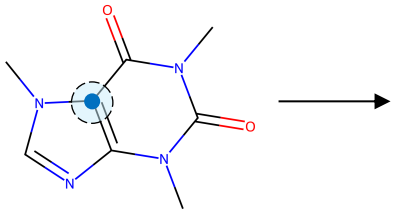
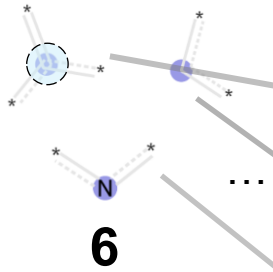
Morgan Fingerprints

Iteration 0: centered on each atom

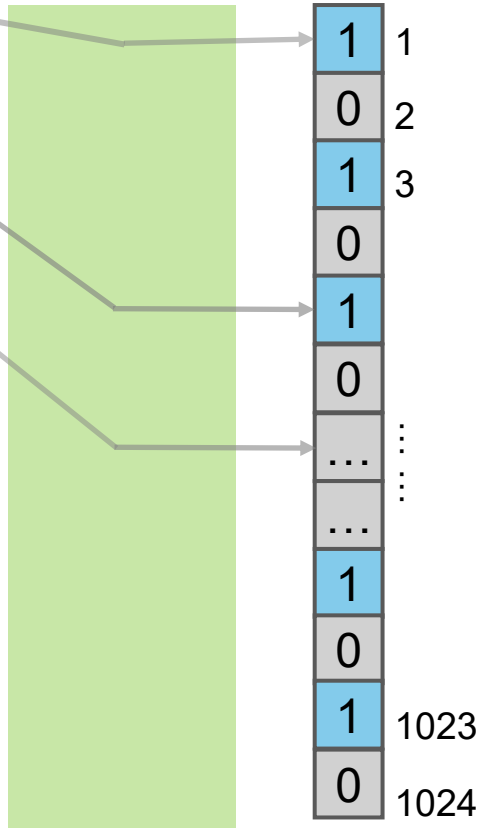


Subgraphs



Hash Function

Fixed-length array



Position in the array

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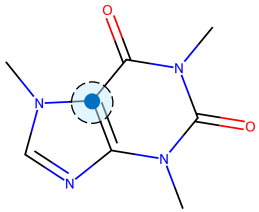
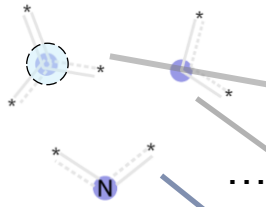
Morgan Fingerprints

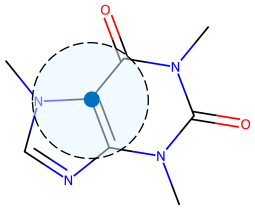
Iteration 0: centered on each atom

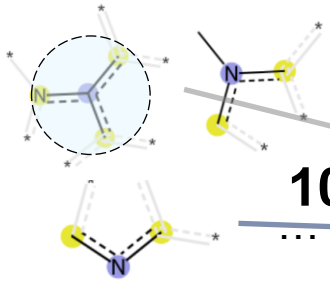




6

Iteration 1: centered on each atom, consider subgraphs extending **one** bond away from a central atom



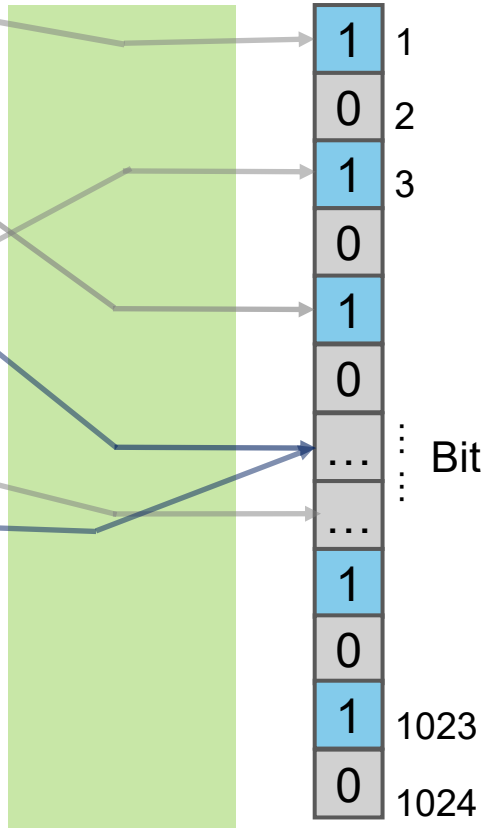


10

Subgraphs

Hash Function

Fixed-length array



Position in the array

Bit collision

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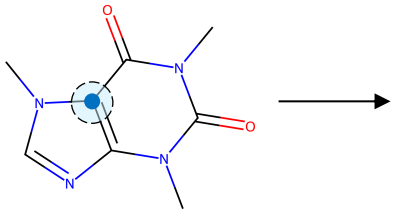
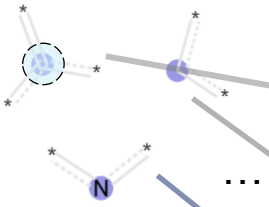
Morgan Fingerprints

Iteration 0: centered on each atom



Subgraphs



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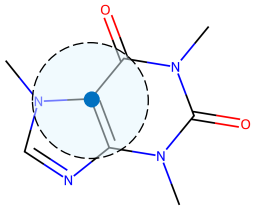
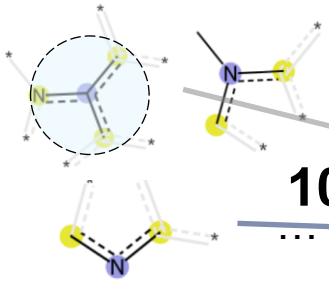
Iteration 1: centered on each atom, consider subgraphs extending **one** bond away from a central atom





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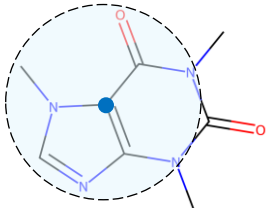
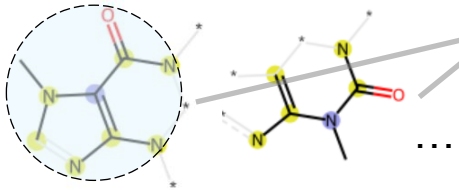
Iteration 2: centered on each atom, consider subgraphs extending **two** bonds away from a central atom

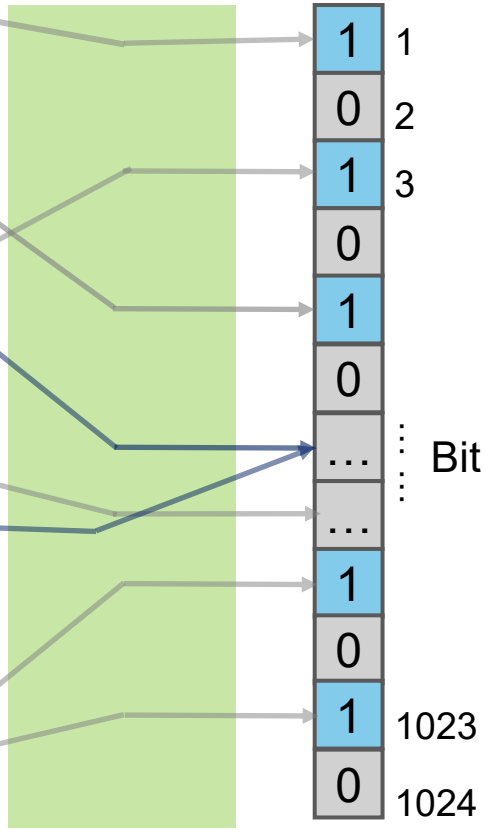




9

Hash Function

Fixed-length array



Position in the array

Bit collision

Total: 25 subgraphs

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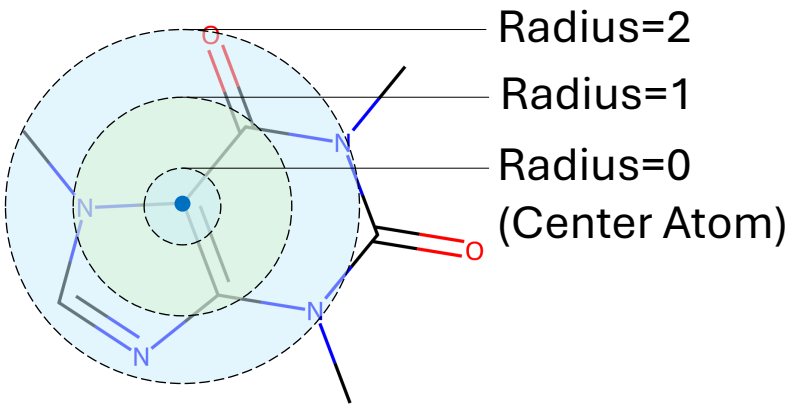
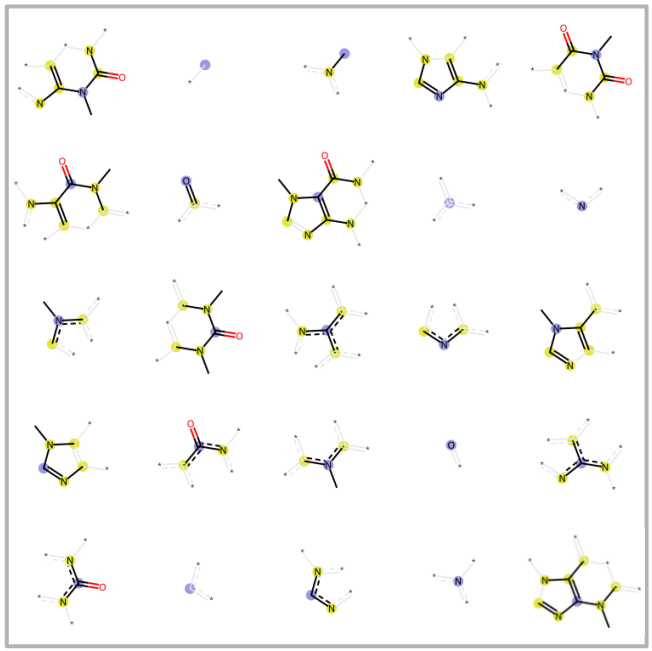
Morgan Fingerprints

Radius: how many bonds away from the center atom:

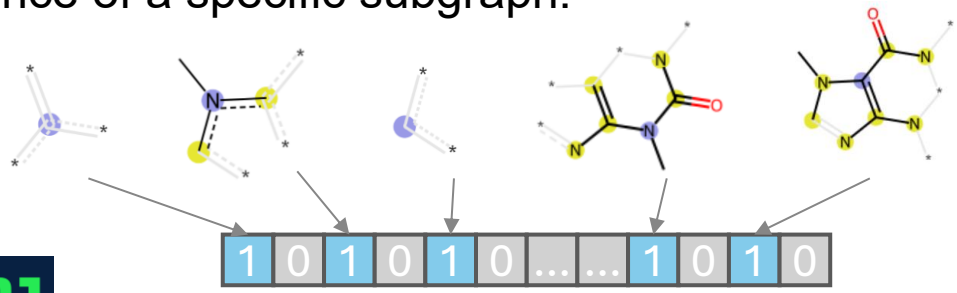


Caffeine

25 subgraphs



Chemical molecules can be represented as binary arrays. Each position indicates the presence or absence of a specific subgraph.



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Morgan Fingerprints

- Morgan fingerprint is a method to compress molecular structure information (subgraphs) to a fixed-length array, with information loss.

- The choice of radius affects how many unique subgraphs can be extracted (often choose 2 or 3).

- When assigning subgraphs to positions in the array (hashing), bit collision may occur where multiple subgraphs are mapped to the same position in the array (choose proper hash size).

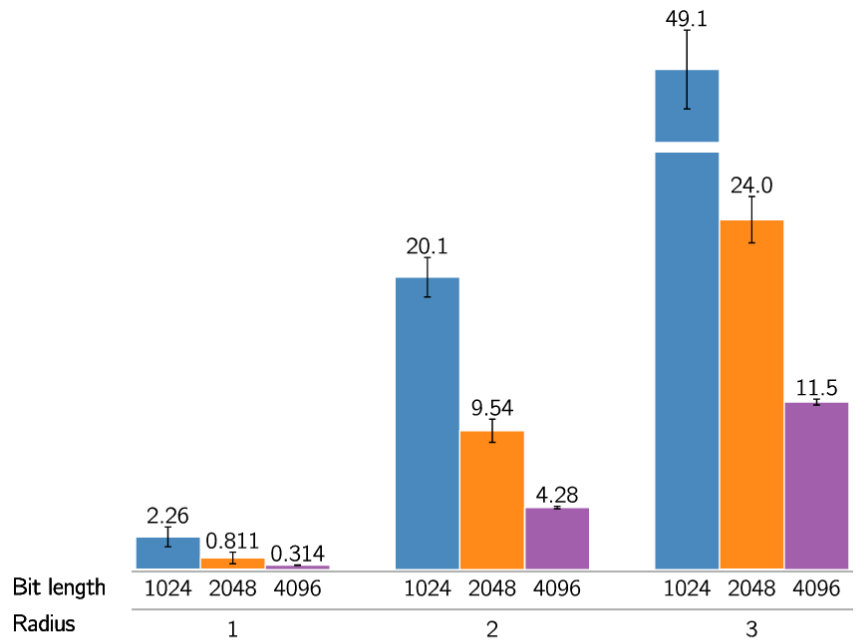

Fig. 1 Collisions per feature. Number of collisions per feature for different hash sizes averaged over the four datasets using the Morgan fingerprint descriptor. Error bars are standard deviation of the mean

J. ChemInform 10, 49, (2018)

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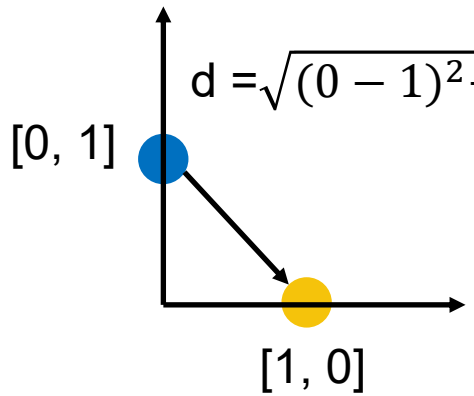
Compound Similarity

Measuring the similarity between pairs of chemical compounds:

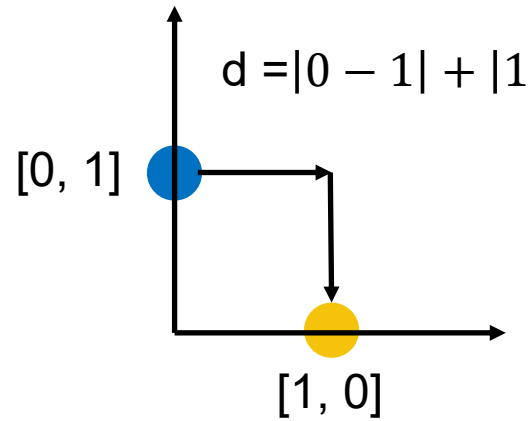
- **Euclidean Distance**

$$d = \sqrt{(0-1)^2 + (1-0)^2} = \sqrt{2}$$



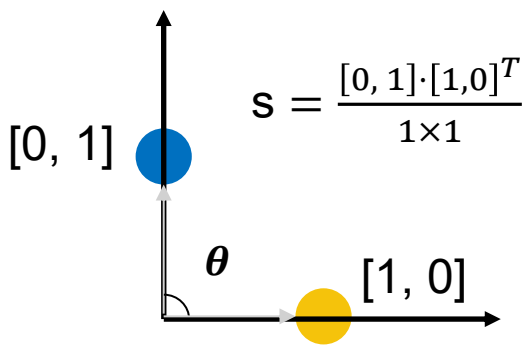
- **Manhattan distance**

$$d = |0-1| + |1-0| = 2$$

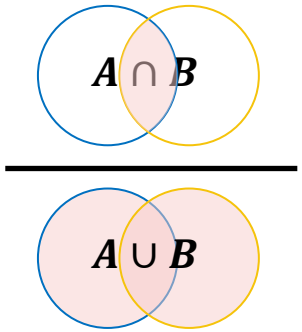


- **Cosine similarity**

$$s = \frac{[0,1]\cdot[1,0]^T}{1\times 1} = 0$$



- **Tanimoto/Jaccard similarity**



$$s = \frac{\sum(0\wedge 1 + 1\wedge 0)}{\sum(0\vee 1 + 1\vee 0)} = 0$$

Two chemical compounds: two arrays, [0,1] & [1,0].

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Outline

- Molecular property descriptors
- **String representations**
- Molecular graphs
- 3D-representations
- Binding environment features
- Bottom-up representations for peptides

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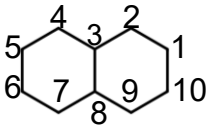
Simplified Molecular Input Line Entry Specification (SMILES)

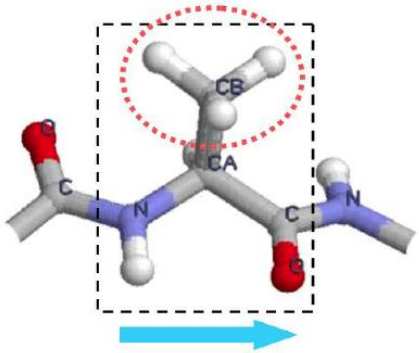
	Rules	**Example**
Atoms	Element Symbol Ions Using brackets	C, N, Cl, … Proton: [H+] Implicit hydrogens can be omitted
Bonds	Single bond: “-” Double bond: “=“ Triple bond: “#” Aromatic bond: “:” or lower case (Single bond and aromatic bond are often omitted)	Benzene: c1ccccc1 or C1=CC=CC=C1 Hydrogen cyanide: C#N
Branches	Put branch in parenthesis after the atom it connects to	Acetic acid: CC(=O)O
Rings	Put the same indices after atoms connected in the ring but not adjacent in the string	 C1CC2CCCCC2CC1

https://www.daylight.com/smiles

© Xuhui Huang

Simplified Molecular Input Line Entry Specification (SMILES)

	Rules	Example
Others	Not connected: “.” Reactions: “>>”	[Na+].[Cl-] C.O=O.O=O>>O=C=O.O.O
Isomeric SMILES: string with information about isotopism, configuration around double bonds and chirality		
Isotopes	Add mass before the atom symbol	Heavy water: [2H]O[2H] Methane [13CH4]
Chirality	Clockwise: “@@” Anti-clockwise: “@”	L-alanine: N[C@@H](C)C(=O)O D-alanine: N[C@H](C)C(=O)O
Double bond configurations	Trans: “/…/” Cis: “/ … \”	Trans: F/C=C/F Cis: F/C=C\F



L-alanine: N[C@@H](C)C(=O)O

From N: (H,methyl,carboxy) appear clockwise.

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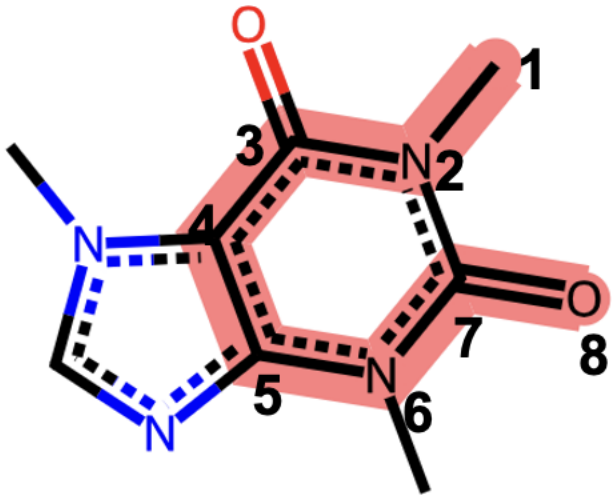
Multiple Representations of SMILES Strings

Cn1c(=O)c2c(ncn2C)n(C)c1=O



Delocalized electron cloud

Caffeine

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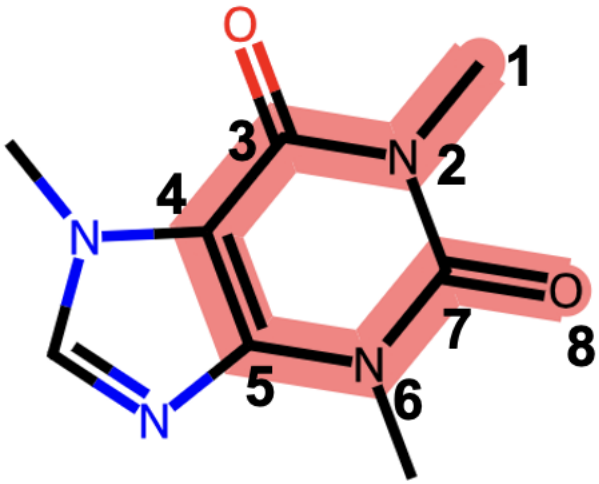
Multiple Representations of SMILES Strings

CN1C(=O)C2=C(N=CN2C)N(C)C1=O



Kekulé Structure

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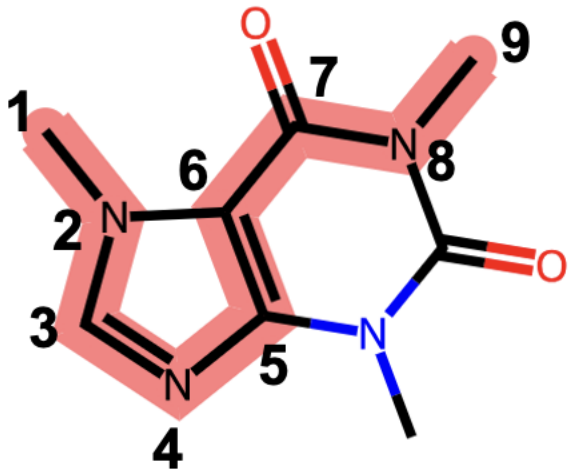
Multiple Representations of SMILES Strings

CN1C=NC2=C1C(=O)N(C(=O)N2C)C



Different choice of the main chain

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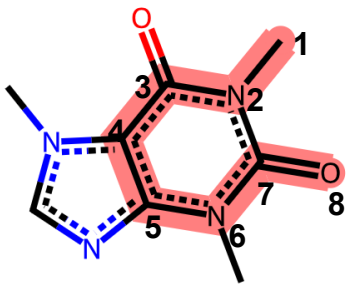
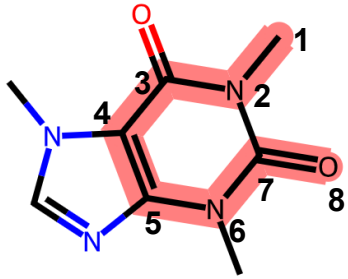
Multiple Representations of SMILES Strings

Cn1c(=O)c2c(ncn2C)n(C)c1=O



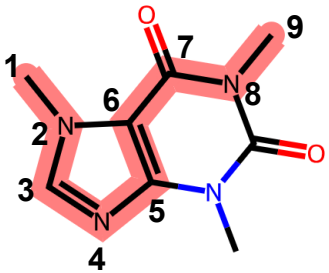
Delocalized electron cloud

CN1C(=O)C2=C(N=CN2C)N(C)C1=O



Kekulé Structure

CN1C=NC2=C1C(=O)N(C(=O)N2C)C



Different choice of the main chain

- A compound could have multiple valid SMILES representations.
- Canonical SMILES has a one-to-one mapping between a compound and its SMILES representation.

J. Chem. Inf. Comput. Sci. 29(2), 97, (1989)

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Exercise

1. Which of the following SMILES strings are valid:

(A). C=CC=C

(B). cccc

(C). C1CCO

(D). C1CC1

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Exercise

1. Which of the following SMILES strings are valid:

(A). C=CC=C - Butadiene.

(B). cccc - Aromaticity cannot be assigned to non-ring molecule

(C). C1CCO - The ring is not closed

(D). C1CC1 - Cyclopropane

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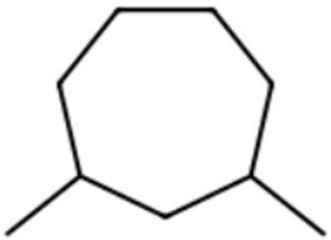
Exercise

2. Which of the following SMILES strings is valid for the chemical compound:



(A). CCc1cccc(C)c1

(B). CCC1CCCC(C1)C

(C). CC1CCCCC(C1)C

(D). CC1CCCC(C)CC1

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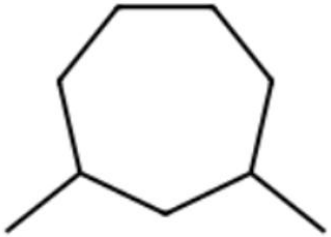
Exercise

2. Which of the following SMILES strings is valid for the chemical compound:



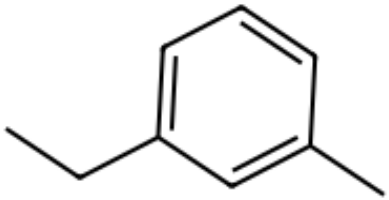
(A). CCc1cccc(C)c1

(B). CCC1CCCC(C1)C

(C). CC1CCCCC(C1)C

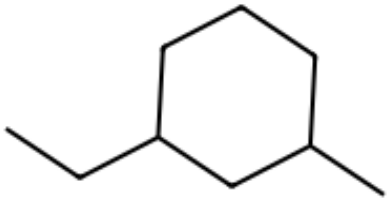
(D). CC1CCCC(C)CC1

A



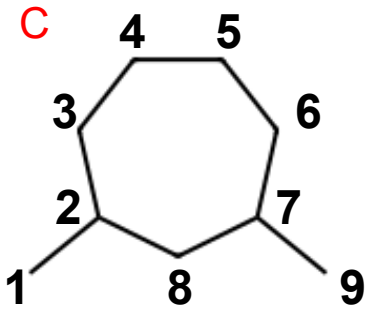
CCc1cccc(C)c1

B



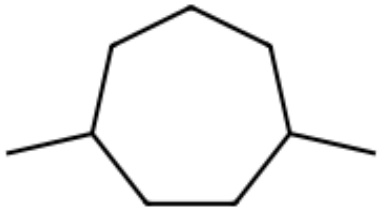
CCC1CCCC(C1)C

C



CC1CCCCC(C1)C

D



CC1CCCC(C)CC1

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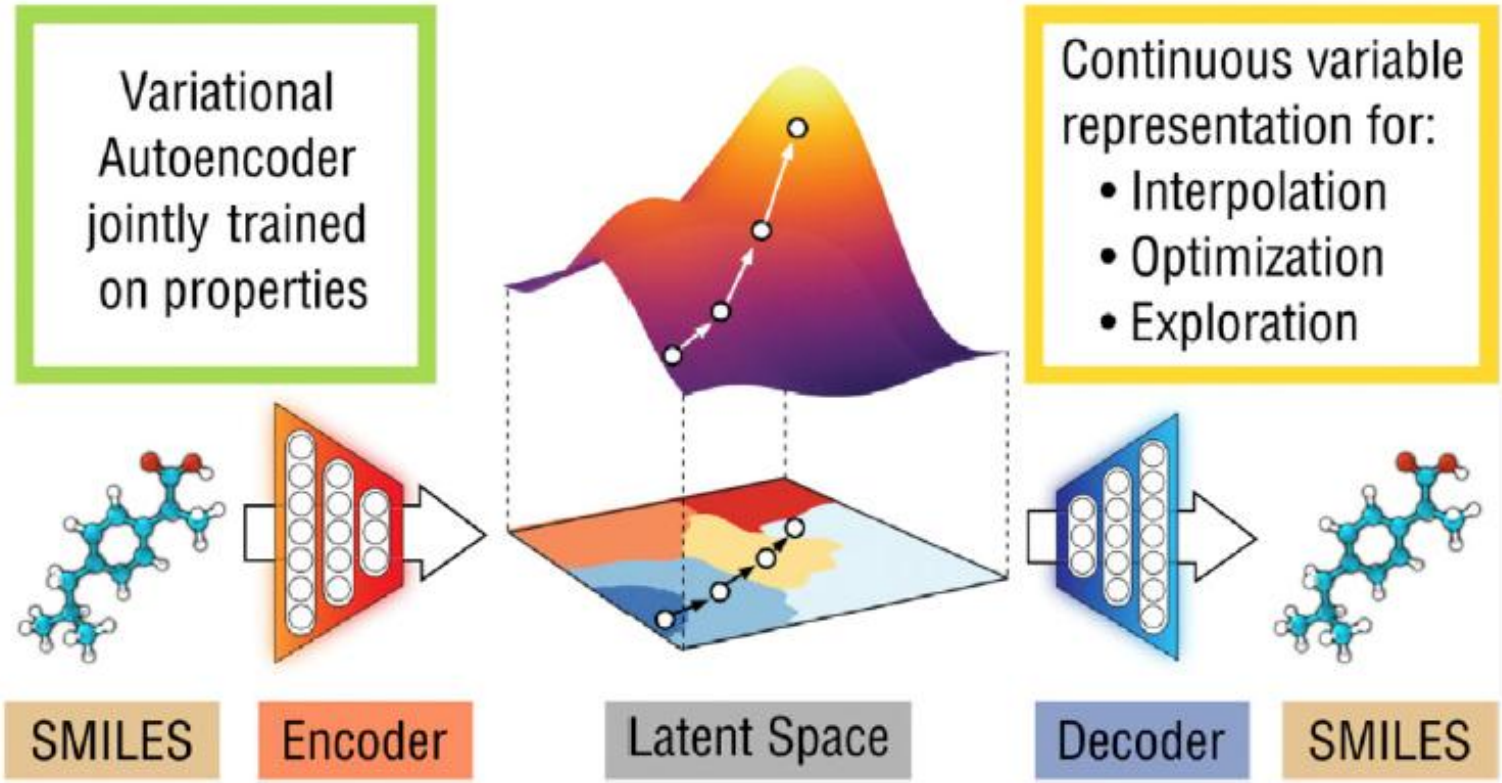
SMILES Provides Input Features to Train ML Models



Variational autoencoder (VAE) trained with joint property prediction to predict novel SMILES strings with high values of desired molecular properties (e.g., logP and QED).

Alán Aspuru-Guzik, *ACS Cent. Sci.* 4(2), 268, (2018)

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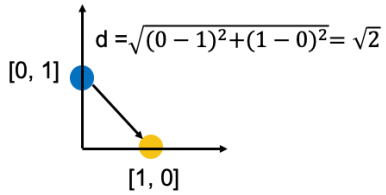
Review of the Previous Lecture

Features of chemical compounds:

- Molecular properties - **drug-likeness molecular properties**: molecular weight (MW), number of hydrogen bond donors (HBD), number of hydrogen bond acceptors (HBA), LogP.
- **Lipinski's rule of five**: MW<= 5, HBD<=5, HBA<=10, LogP<=5.0. Most drugs violate 0 or 1 rule.
- **Fingerprint** of a chemical compound: an array storing pre-defining a list of properties.
- **Morgan Fingerprint**: compress molecular structure information into subgraphs to store in a fixed-length array, with information loss.
  - Number of radius – often 2 or 3
  - Size of the fixed-length array: 1024-bit, 2048-bit…
  - Bit collision - multiple subgraphs are mapped to the same position in the array
- **Similarity measure**: Euclidean distance, Manhattan distance, cosine similarity, Tanimoto/Jaccard similarity.



Euclidean distance

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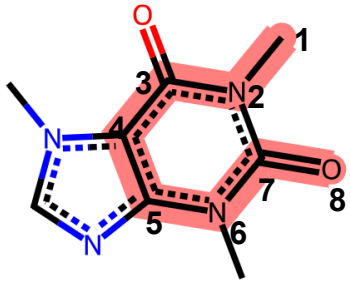
Review of the Previous Lecture

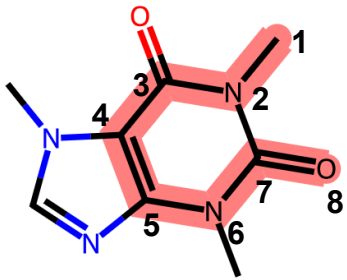
SMILES string representations of chemical compounds:

Cn1c(=O)c2c(ncn2C)n(C)c1=O



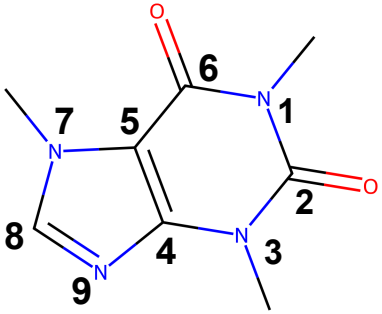
Canonical SMILES in **RDKit**

CN1C(=O)C2=C(N=CN2C)N(C)C1=O



Canonical SMILES in **PubChem**

1,3,7-Trimethylpurine-2,6-dione



IUPAC name

- Canonical SMILES has a one-to-one mapping between a compound and its SMILES representation – based on molecular graphs.

J. Chem. Inf. Comput. Sci. 29(2), 97, (1989)

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Outline

- Molecular property descriptors
- String representations
- **Molecular graphs**
- 3D-representations
- Binding environment features
- Bottom-up representations for peptides

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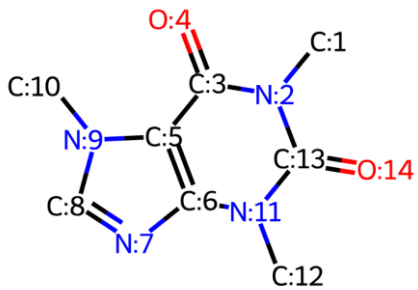
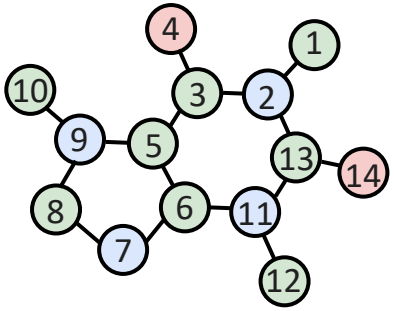
Molecular Graphs

- Atoms as Nodes: **V** - One-hot Encoding (no hydrogen atoms).
- Bonds as edges: **E** - Adjacency Matrix.

$A[i,j]$ represents if atom i and j are connected by chemical bond.





G = (V, E)

Node Features
One-hot Encoding

$H^0 =$

	C	N	O	S	P
1	■				
2		■			
3	■				
4			■		
5	■				
6	■				
7		■			
8	■				
9	■				
10	■				
11		■			
12	■				
13	■				
14			■		

14×5

Adjacency Matrix

$A =$

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1		■												
2	■		■										■	
3		■		■	■									
4			■											
5			■			■			■					
6					■		■				■			
7						■		■						
8							■		■					
9					■			■		■				
10									■					
11						■						■	■	
12											■			
13		■									■			■
14													■	

14×14

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Molecular Graphs

Adjacency Matrix

$A=$

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1		■												
2	■		■										■	
3		■		■	■									
4			■											
5			■			■			■					
6					■		■				■			
7						■		■						
8							■		■					
9					■			■		■				
10									■					
11						■						■	■	
12											■			
13		■									■			■
14													■	

14×14

Coordinate (COO) matrix

	Source node	Target node
1	1	2
2	2	1
3	2	3
4	3	2
⋮	⋮	⋮
⋮	⋮	⋮
29	13	14
30	14	13

30×2

30 chemical bonds in Caffeine.

Edge Features One-hot Encoding

	-	=	#	:
1	■			
2	■			
3				■
4				■
⋮		■		
⋮	■			
29	■			
30			■	

30×4

Single bond “-”, double bond “=“, triple bond “#”, aromatic bond: “:”

COO matrix is more efficient than adjacency matrix in storage!

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Outline

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- Bottom-up representations for peptides

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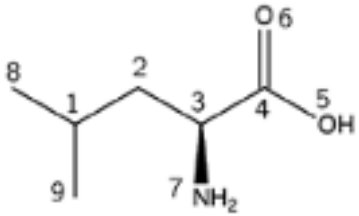
Cartesian Coordinates

MDL molfile:

Leucine



Chiral

```
  9  8  0  0  1  0  0  0  0  0999 V2000
   -1.0717   -0.2062    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
   -0.3572    0.2062    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
    0.3572   -0.2062    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
    1.0717    0.2062    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
    1.7861   -0.2062    0.0000 O   0  0  0  0  0  0  0  0  0  0  0  0
    1.0717    1.0312    0.0000 O   0  0  0  0  0  0  0  0  0  0  0  0
    0.3572   -1.0312    0.0000 N   0  0  0  0  0  0  0  0  0  0  0  0
   -1.7861    0.2062    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
   -1.0717   -1.0312    0.0000 C   0  0  0  0  0  0  0  0  0  0  0  0
  1  2  1  0
  2  3  1  0
  3  4  1  0
  4  5  1  0
  4  6  2  0
  3  7  1  1
  1  8  1  0
  1  9  1  0
M  END
```

Blocks not used in this Ctab:
Atom List, Stext, Properties

Counts Line

Atom Block

Bond Block

M END

Connection Table (Ctab)

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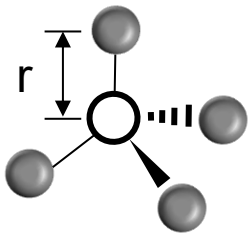
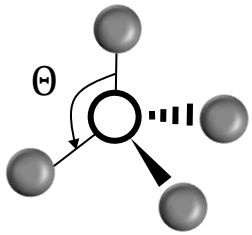
Internal Coordinates

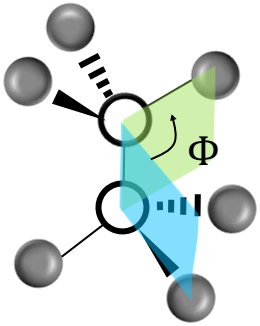
Bond length



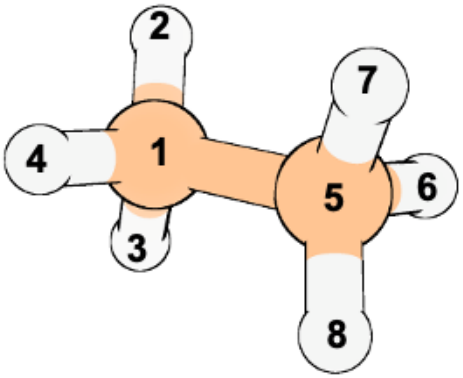
Bond angle



Torsion angle



Example:



Topology file: “.top”

```
[ bonds ]
;   ai     aj  funct    r            k
     1      2    1    1.0970e-01   2.7665e+05 ;     C1 - H1
     1      3    1    1.0970e-01   2.7665e+05 ;     C1 - H2
     1      4    1    1.0970e-01   2.7665e+05 ;     C1 - H3
     1      5    1    1.5380e-01   2.5179e+05 ;     C1 - C2
     5      6    1    1.0970e-01   2.7665e+05 ;     C2 - H4
     5      7    1    1.0970e-01   2.7665e+05 ;     C2 - H5
     5      8    1    1.0970e-01   2.7665e+05 ;     C2 - H6
```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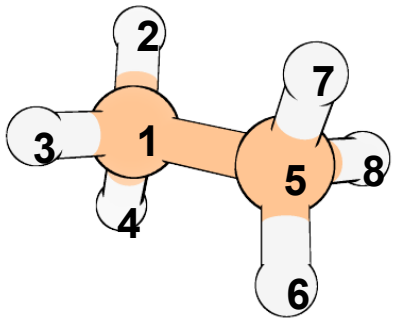
Internal Coordinates

Topology file: “.top”




```
[ angles ]
;   ai     aj     ak    funct   theta        cth
     1      5      6      1    1.0980e+02   3.8744e+02 ;    C1 - C2     - H4
     1      5      7      1    1.0980e+02   3.8744e+02 ;    C1 - C2     - H5
     1      5      8      1    1.0980e+02   3.8744e+02 ;    C1 - C2     - H6
     2      1      3      1    1.0758e+02   3.2970e+02 ;    H1 - C1     - H2
     2      1      4      1    1.0758e+02   3.2970e+02 ;    H1 - C1     - H3
     2      1      5      1    1.0980e+02   3.8744e+02 ;    H1 - C1     - C2
     3      1      4      1    1.0758e+02   3.2970e+02 ;    H2 - C1     - H3
     3      1      5      1    1.0980e+02   3.8744e+02 ;    H2 - C1     - C2
     4      1      5      1    1.0980e+02   3.8744e+02 ;    H3 - C1     - C2
     6      5      7      1    1.0758e+02   3.2970e+02 ;    H4 - C2     - H5
     6      5      8      1    1.0758e+02   3.2970e+02 ;    H4 - C2     - H6
     7      5      8      1    1.0758e+02   3.2970e+02 ;    H5 - C2     - H6
```

```
[ dihedrals ] ; propers
; for gromacs 4.5 or higher, using funct 9
;    i      j      k      l    func   phase     kd      pn
     2      1      5      6      9   300.00   0.62760   3 ;    H1-    C1-    C2-    H4
     2      1      5      7      9    60.00   0.62760   3 ;    H1-    C1-    C2-    H5
     2      1      5      8      9   180.00   0.62760   3 ;    H1-    C1-    C2-    H6
     3      1      5      6      9    60.00   0.62760   3 ;    H2-    C1-    C2-    H4
     3      1      5      7      9   180.00   0.62760   3 ;    H2-    C1-    C2-    H5
     3      1      5      8      9   300.00   0.62760   3 ;    H2-    C1-    C2-    H6
     4      1      5      6      9   180.00   0.62760   3 ;    H3-    C1-    C2-    H4
     4      1      5      7      9   300.00   0.62760   3 ;    H3-    C1-    C2-    H5
     4      1      5      8      9    60.00   0.62760   3 ;    H3-    C1-    C2-    H6
```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Databases with 3D Coordinates

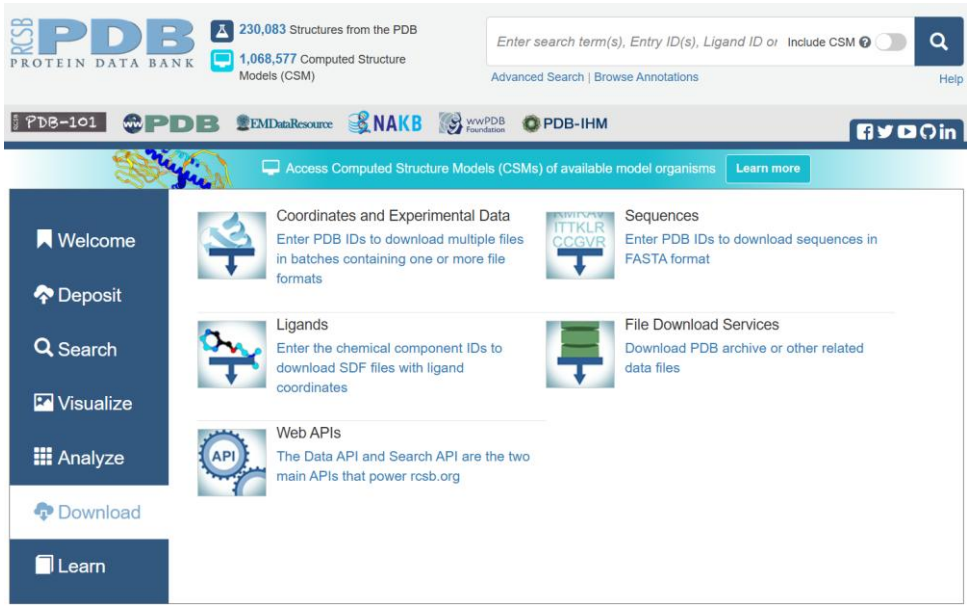
PubChem, ZINC, RSCB PDB and RDKit all contain 3D coordinates of compounds

ZINC20

Welcome to ZINC, a free database of commercially-available compounds for virtual screening. ZINC contains over 230 million purchasable compounds in ready-to-dock, 3D formats. ZINC also contains over 750 million purchasable compounds you can search for analogs in under a minute.

230 million compounds in 3D format.

https://zinc20.docking.org/



RCSB PDB contains

- >200 thousand experimentally determined bio-macromolecule structures
- >1million computed structure models (i.e. homology modeling, AlphaFold generations)
- Ligand coordinates

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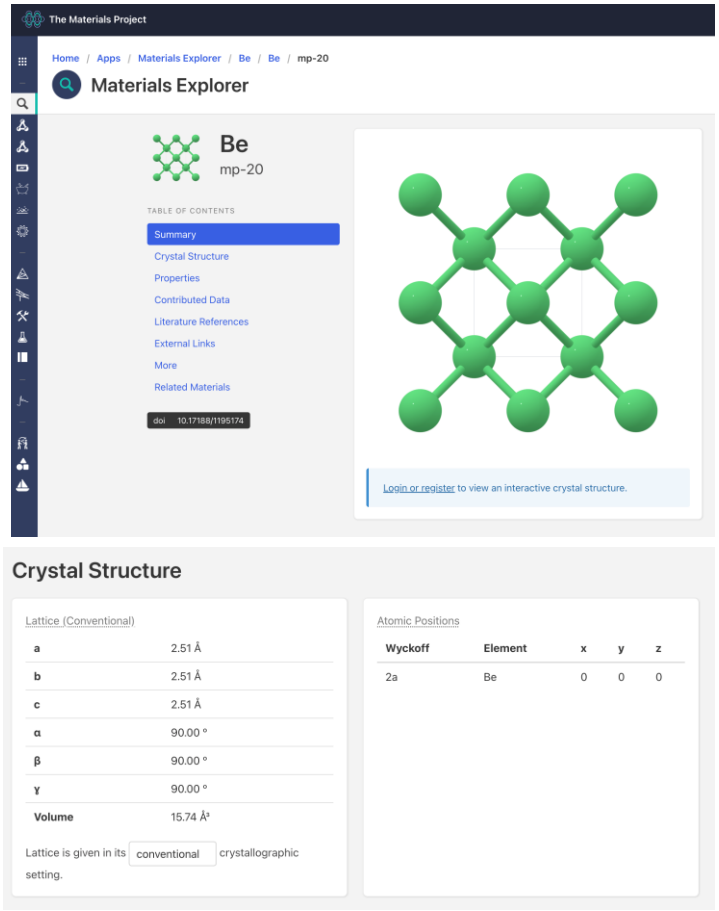
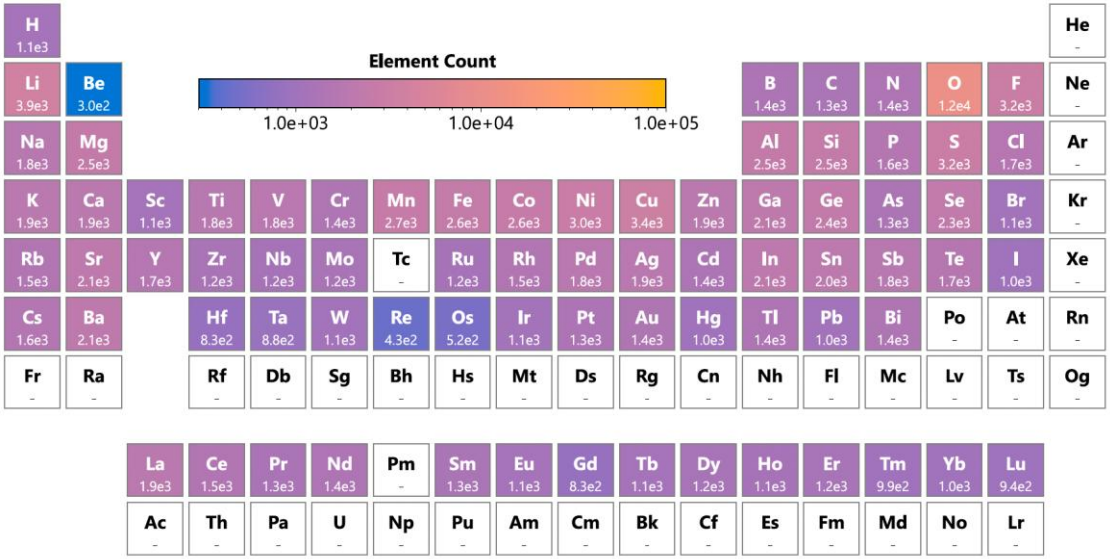
Databases with 3D Coordinates

Materials Project (MP)-20 contains crystal structures of inorganic materials:





(a) MP-20

DFT-relaxed crystal structures obtained from experimentally-known crystals and other resources.

https://next-gen.materialsproject.org/materials/mp-20

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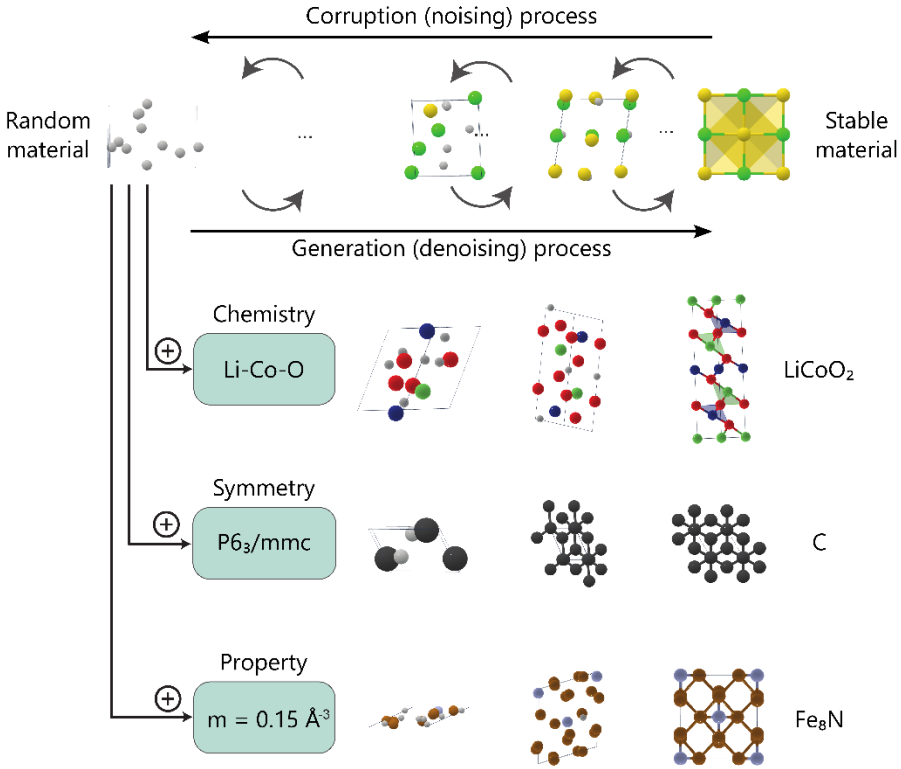
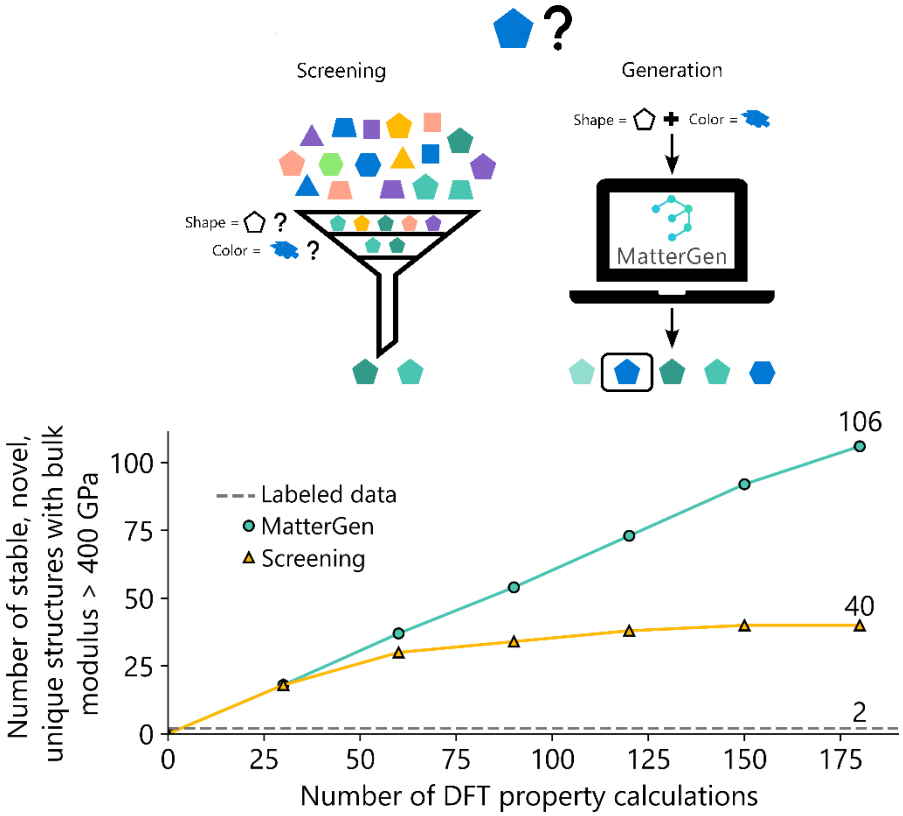
A Generative Model for Inorganic Materials Design

Data: 850,384 unique and stable structures of inorganic materials.





MatterGen: A diffusion model (ch.8) that operates on the 3D geometry of materials.

MatterGen outperforms screening when fine tuned to predict materials with high bulk modulus.

Nature (2025), 01/16/2025

https://www.nature.com/articles/s41586-025-08628-5

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Outline

- Molecular property descriptors
- String representations
- Molecular graphs
- 3D-representations
- **Binding environment features**
- Bottom-up representations for peptides

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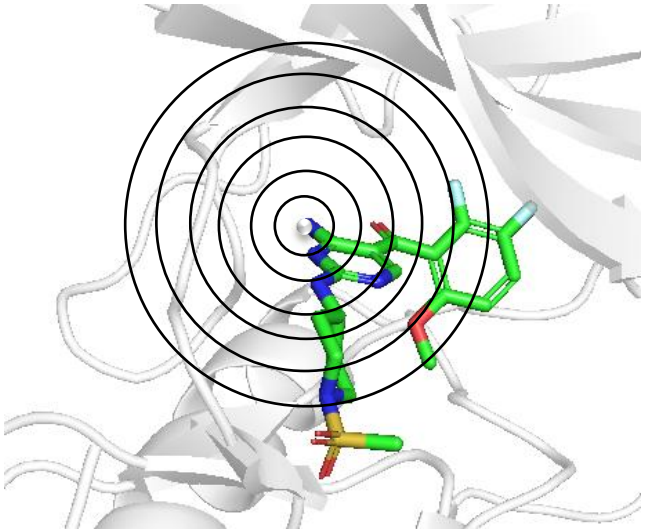
Binding Environment Features

FEATURE vectors capture the physicochemical properties of the ligand binding site: atom type, residue (e.g., name, hydrophobic/polar, charge), secondary structure.



Atom-type

ATOM-TYPE-IS-C
ATOM-TYPE-IS-CT
ATOM-TYPE-IS-Ca
ATOM-TYPE-IS-N
ATOM-TYPE-IS-N2
ATOM-TYPE-IS-N3
ATOM-TYPE-IS-Na
ATOM-TYPE-IS-O
ATOM-TYPE-IS-O2
ATOM-TYPE-IS-OH
ATOM-TYPE-IS-S
ATOM-TYPE-IS-SH
ATOM-TYPE-IS-OTHER
ATOM-NAME-IS-ANY
ATOM-NAME-IS-C
ATOM-NAME-IS-N
ATOM-NAME-IS-O
ATOM-NAME-IS-S
ATOM-NAME-IS-OTHER
HYDROXYL
AMIDE
AMINE
CARBONYL
RING-SYSTEM
PEPTIDE

Secondary-Structure

SECONDARY_STRUCTURE1_IS_3HELIX
SECONDARY_STRUCTURE1_IS_4HELIX
SECONDARY_STRUCTURE1_IS_5HELIX
SECONDARY_STRUCTURE1_IS_BRIDGE
SECONDARY_STRUCTURE1_IS_STRAND
SECONDARY_STRUCTURE1_IS_TURN
SECONDARY_STRUCTURE1_IS_BEND
SECONDARY_STRUCTURE1_IS_COIL
SECONDARY_STRUCTURE1_IS_HET
SECONDARY_STRUCTURE1_IS_UNKNOWN
SECONDARY_STRUCTURE2_IS_HELIX
SECONDARY_STRUCTURE2_IS_BETA
SECONDARY_STRUCTURE2_IS_COIL
SECONDARY_STRUCTURE2_IS_HET
SECONDARY_STRUCTURE2_IS_UNKNOWN

Residue

RESIDUE_NAME_IS_ALA
RESIDUE_NAME_IS_ARG
RESIDUE_NAME_IS_ASN
RESIDUE_NAME_IS_ASP
RESIDUE_NAME_IS_CYS
RESIDUE_NAME_IS_GLN
RESIDUE_NAME_IS_GLU
RESIDUE_NAME_IS_GLY
RESIDUE_NAME_IS_HIS
RESIDUE_NAME_IS_ILE
RESIDUE_NAME_IS_LEU
RESIDUE_NAME_IS_LYS
RESIDUE_NAME_IS_MET
RESIDUE_NAME_IS_PHE
RESIDUE_NAME_IS_PRO
RESIDUE_NAME_IS_SER
RESIDUE_NAME_IS_THR
RESIDUE_NAME_IS_TRP
RESIDUE_NAME_IS_TYR
RESIDUE_NAME_IS_VAL
RESIDUE_NAME_IS_HOH
RESIDUE_NAME_IS_OTHER
CLASS1_IS_HYDROPHOBIC
CLASS1_IS_CHARGED
CLASS1_IS_POLAR
CLASS1_IS_UNKNOWN
CLASS2_IS_NONPOLAR
CLASS2_IS_POLAR
CLASS2_IS_BASIC
CLASS2_IS_ACIDIC
CLASS2_IS_UNKNOWN
PARTIAL-CHARGE
VDW-VOLUME
CHARGE
CHARGE-WITH-HIS
NEG-CHARGE
POS-CHARGE
HYDROPHOBICITY
MOBILITY
SOLVENT-ACCESSIBILITY

80 physicochemical properties × 6 concentric shells = 480 features

BMC Genomics 9, S2 (2008)

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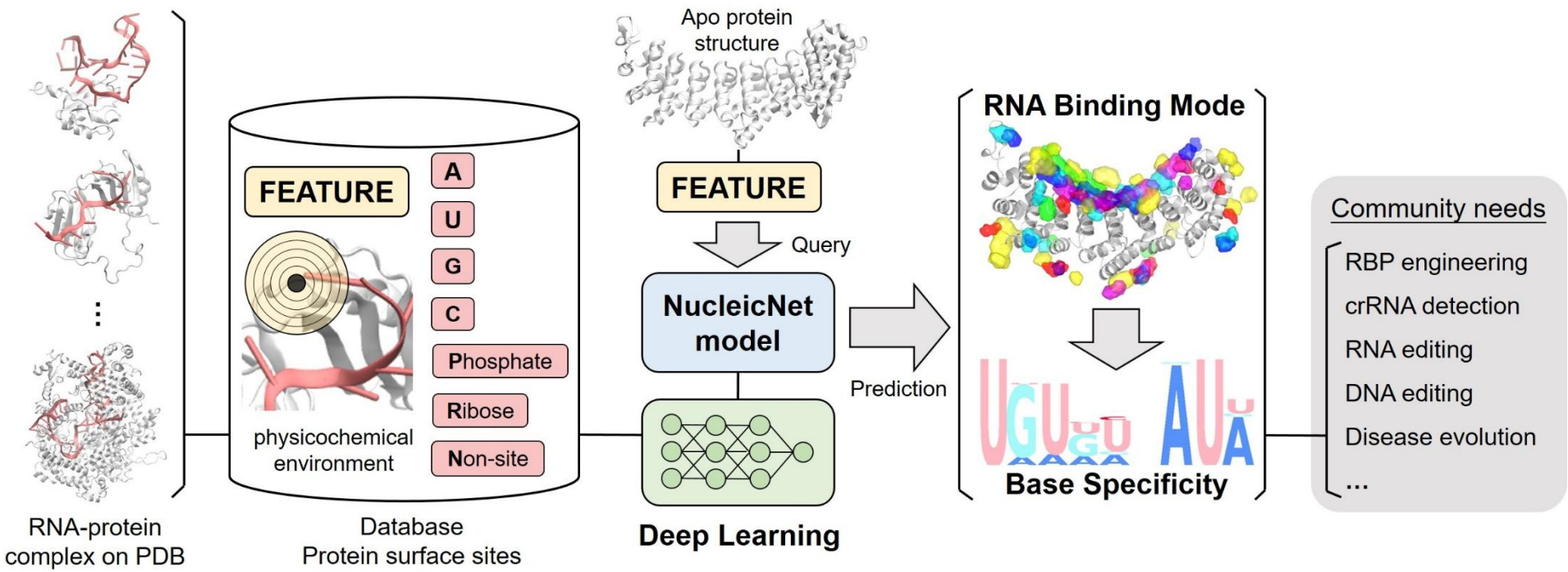
Predict Protein-RNA Interactions Based on Binding Environment Features

Data: Protein-RNA complex structures from PDB data bank.



FEATURE — Atom-type, Residue (name, hydrophobic/polar, charge, …), Secondary structure (3-helix, 4-helix, bridge, strand, turn, band, coil, …)

NucleicNet: A Convolutional Neural Network (Ch. 6) platform to predict protein-RNA interactions.

Nature Communications, 10: 4941, (2019)

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Outline

- Molecular property descriptors
- String representations
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- 3D-representations
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- **Bottom-up representations for peptides**

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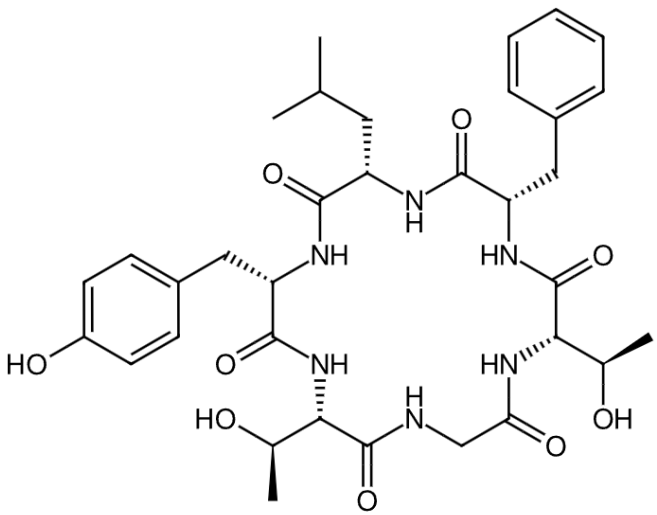
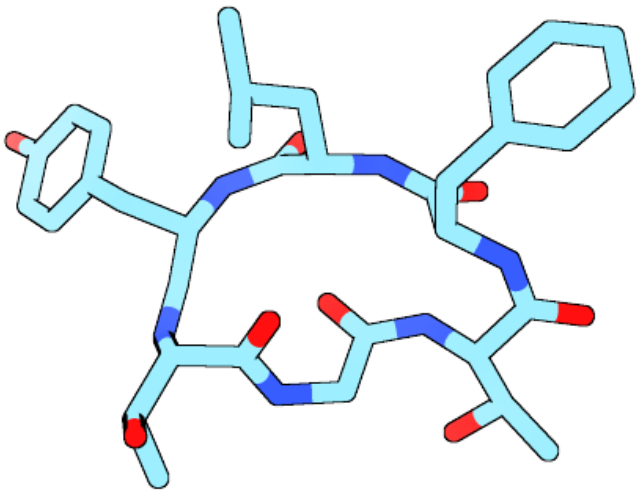
SMILES Strings Applied to Peptides

A 6-residue cyclic peptide:





Sequence:

{G.T.F.L.Y.T}

SMILES (118 characters)

CC(C)C[C@@H]1NC(=O)[C@H](Cc2ccccc2)NC(=O)[C@H]([C@@H](C)O)NC(=O)CN
C(=O)[C@H]([C@@H](C)O)NC(=O)[C@H](Cc2ccc(O)cc2)NC1=O

http://cycpeptmpdb.com/

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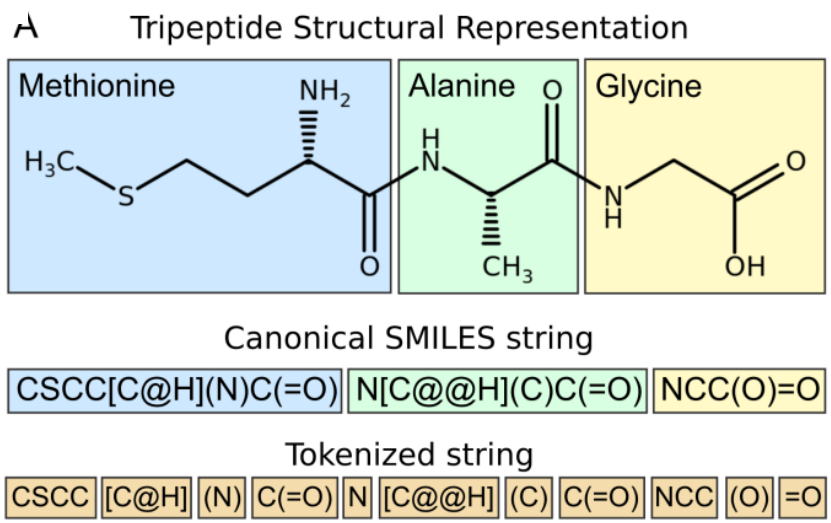
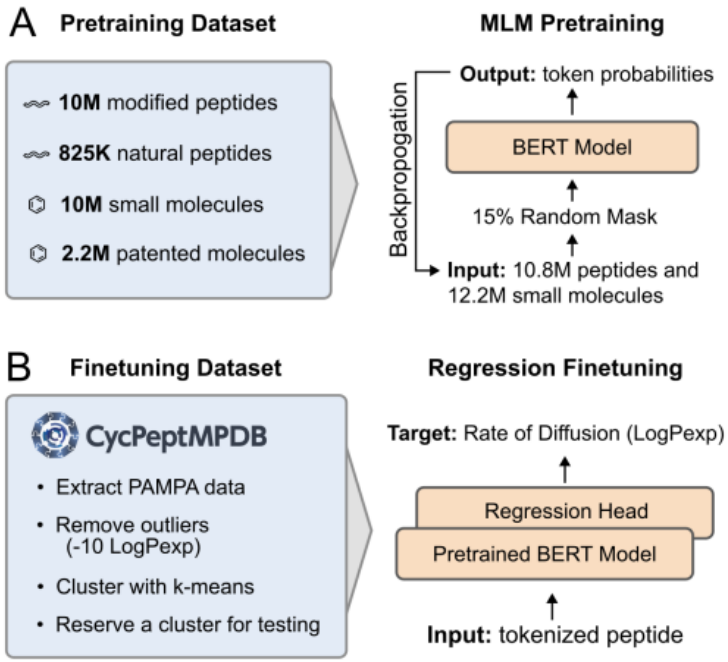
Large Language Models for Peptides

The chemical large language model, PeptideCLM, is trained on SMILES strings of peptides and small molecules.





Training Data (Size)	RMSE
Full (23M)	0.742 ± 0.214
Peptide (11M)	0.723 ± 0.175
Chemical (12M)	0.839 ± 0.263

PeptideCLM is fined tuned (against 7,000 experimental data) to predict membrane permeation for cyclic peptides

JCIM, (2025), Jan 8, 2025
DOI:10.1021/acs.jcim.4c01441

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Summary

	Molecular Descriptors & Fingerprints	SMILES	Molecular Graphs	3D Coordinates
Format	Numeric array	String	Matrices	Files
Information Encoded	Structural patterns	Atoms, Bonds, Connectivity	Atoms, Bonds, Connectivity, sometimes conformation	Atoms, Bonds, Angles, Torsions, Connectivity, Conformation
Storage Size	Small	Small	Large	Large
Human-readable	No	Yes	Yes	Yes
Applications in ML methods	Input features for various ML methods	Often used in natural language processing (NLP) models	Often used in graph neural networks (GNN)	Requires preprocessing and equivariant neural networks

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